East West University

Department of Computer Science and Engineering

University Campus Network Using Cisco Packet Tracer

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Abstract

Networking has become the backbone of communication, enabling organizations to operate effectively. This project focuses on designing an **Enhanced University Network** using Cisco Packet Tracer. The objective is to create a secure and efficient network architecture that integrates dynamic IP allocation, VLAN segmentation, and robust routing protocols.

This work simulates a university campus network comprising multiple departments and external connectivity to cloud-hosted servers. Advanced security configurations ensure protection against common threats, while protocols like OSPF and DHCP enhance network efficiency. The project's findings emphasize the importance of topology planning and network simulation for educational institutions.

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Chapter 1: Introduction

1.1 Objectives

1. To design a secure and scalable network for a university campus.
2. To simulate the network using Cisco Packet Tracer.
3. To integrate dynamic IP addressing and VLAN segmentation.

1.2 Overview

In today's digital era, efficient networking is critical for educational institutions. This project presents a detailed simulation of a university network, focusing on connectivity, security, and cost-effectiveness.

1.3 Motivation

With the increasing reliance on digital resources, universities must adopt modern network designs to enhance communication and resource-sharing. This project aims to address these needs by proposing a robust and flexible network infrastructure.

1.4 Report Structure

Chapter 2 discusses the existing literature on network design. Chapter 3 outlines the proposed methodology, including the network diagram and protocols used. Chapter 4 presents the simulation results, and Chapter 5 concludes the report with future recommendations.

Chapter 2: Proposed Methodology

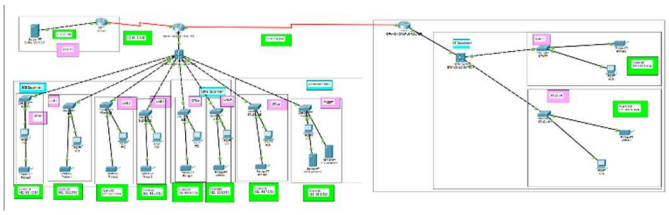
2.1 System Requirements

The network includes the following components:

- Cisco 2911 Routers
- Layer 3 Switches
- VLAN-enabled Layer 2 Switches
- Dynamic Host Configuration Protocol (DHCP) servers

2.2 Network Diagram

The proposed topology connects four departments: Electrical Engineering, Computer Science, Mathematics, and the Admissions Office. Each department operates on separate VLANs, ensuring logical segregation.



2.3 Protocols Used

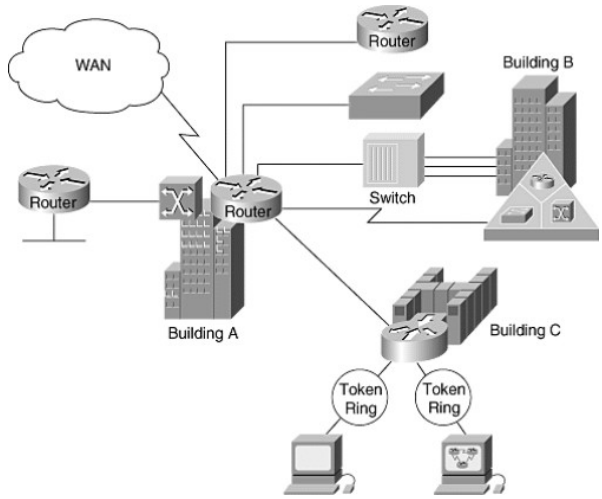
- **OSPF:** For dynamic routing.
- **DHCP:** For automatic IP allocation.
- **ACLs:** For enhanced security.

2.4 Configuration Details

The network is configured using Cisco Packet Tracer. Each VLAN is assigned a unique subnet, and routers are programmed for inter-VLAN

routing. Security measures include access control lists (ACLs) and port security.

2.5 System Models



Chapter 3: Simulation and Results

3.1 Logical Topology

The logical topology demonstrates interconnectivity among departments and external servers. Packet Tracer simulations confirm successful data transmission between nodes.

3.2 Simulation:

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC5	ICMP
	0.000	--	PC5	ARP
	0.001	PC5	A&D	ARP
	0.002	A&D	MAIN-CAMPUS L3 SWITCH	ARP
	0.002	A&D	Printer5	ARP
	0.003	MAIN-CAMPUS L3 SWITCH	MAIN-CAMPUS ROUTER	ARP
	0.004	MAIN-CAMPUS ROUTER	MAIN-CAMPUS L3 SWITCH	ARP
	0.005	MAIN-CAMPUS L3 SWITCH	A&D	ARP
	0.006	A&D	PC5	ARP
	0.006	--	PC5	ICMP
	0.007	PC5	A&D	ICMP
	0.008	A&D	MAIN-CAMPUS L3 SWITCH	ICMP
	0.009	MAIN-CAMPUS L3 SWITCH	MAIN-CAMPUS ROUTER	ICMP
	0.009	--	MAIN-CAMPUS ROUTER	ARP
	0.010	MAIN-CAMPUS ROUTER	MAIN-CAMPUS L3 SWITCH	ARP
	0.011	MAIN-CAMPUS L3 SWITCH	STUD-LAB	ARP
	0.012	STUD-LAB	PC6	ARP
	0.012	STUD-LAB	Printer6	ARP
	0.013	PC6	STUD-LAB	ARP
	0.014	STUD-LAB	MAIN-CAMPUS L3 SWITCH	ARP
	0.015	MAIN-CAMPUS L3 SWITCH	MAIN-CAMPUS ROUTER	ARP

Using Command Prompt

The ping command is a very common method for troubleshooting the accessibility of devices. It uses a series of Internet Control Message Protocol (ICMP) Echo messages to determine:

- Whether a remote host is active or inactive.
- The round-trip delay in communicating with the host.
- Packet loss.
- The ping command first sends an echo request packet to an address, then waits for a reply. The ping is successful only if:
  - The echo request gets to the destination, and the destination can get an echo reply to the source within a predetermined time called a timeout.
  - The default value of this timeout is two seconds on Cisco routers.

Fig. 1 and 2 show a different way to check if our network connectivity is reliable. We have used command prompt of PC2 and have ping two hosts: one from the branch/CS Department host i.e., PC8 with IP address 192.168.10.2/24 and second from Admission Department host's i.e., PC6 with IP address 192.168.7.2/24.

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=2ms TTL=126
Reply from 192.168.10.2: bytes=32 time=9ms TTL=126
Reply from 192.168.10.2: bytes=32 time=5ms TTL=126
Reply from 192.168.10.2: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 9ms, Average = 6ms
```

Figure 1 : Ping from PC2 to PC8

```
C:\>ping 192.168.7.2

Pinging 192.168.7.2 with 32 bytes of data:

Reply from 192.168.7.2: bytes=32 time<1ms TTL=127
Reply from 192.168.7.2: bytes=32 time<1ms TTL=127
Reply from 192.168.7.2: bytes=32 time<1ms TTL=127
Reply from 192.168.7.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.7.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 2 Ping from PC2 to PC8

3.4 Performance Analysis

Performance metrics include latency, packet delivery success rate, and bandwidth utilization. The network achieves optimal results, with minimal packet loss and efficient resource allocation.

Chapter 4: Conclusion and Future Work

4.1 Summary of Findings

The project successfully demonstrates a secure and scalable university network using Cisco Packet Tracer. Key achievements include VLAN segmentation, dynamic IP allocation, and secure inter-department communication.

4.2 Recommendations

Future work can focus on:

1. Practical implementation of the proposed design.
2. Enhancing network security with advanced firewalls.
3. Extending the design to support wireless connectivity.

Bibliography

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